Cut Locus of Submanifolds:
Stability and Intersection with Focal Locus

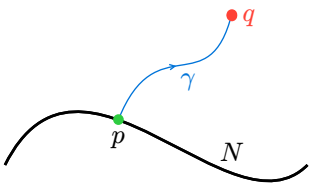
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Introduction

Given a submanifold N of a Riemannian manifold $(M, \mathfrak{g})$, and a point $q \in M$, an N-segment joining N to q is a unit-speed geodesic $\gamma : [0, L] \to M$, such that

$$p := \gamma(0) \in N, \quad \gamma(L) = q, \quad \mathrm{Len}(\gamma) = L = \mathrm{dist}_{\mathfrak{g}}(N, q).$$



Definition

The cut locus $\mathrm{Cut}(N)$ of a submanifold $N \subset M$ is the set of points $q \in M$ such that there exists an N-segment joining N to q, any extension of which fails to be distance-minimizing from N.

$$M \setminus \mathrm{Cut}(N) \underset{\text{s.d.r}}{\simeq} N, \qquad M \setminus N \underset{\text{s.d.r}}{\simeq} \mathrm{Cut}(N).$$

Definition

Given $N \subset M$, the separating set of N is the collection

$$\mathrm{Sep}(N) := \{q \in M \mid \text{There exist at least two distinct } N\text{-segments joining } N \text{ to } q\}$$

Theorem : (_, Prasad [2024][†])

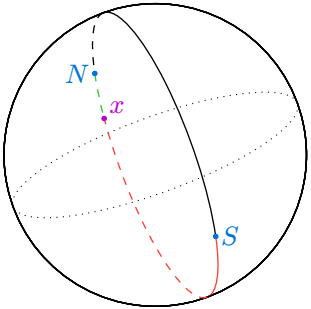
$$\mathrm{Cut}(N) = \overline{\mathrm{Sep}(N)}.$$

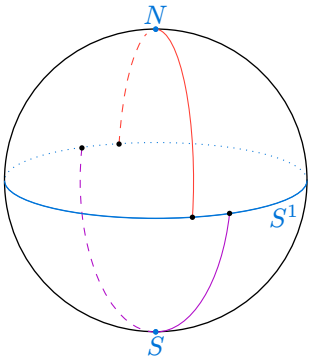
[†] **Aritra Bhowmick** and Sachchidanand Prasad. *On the cut locus of submanifolds of a Finsler manifold.* J. Geom. Anal. (2024).

Examples

The cut locus of the south pole S on the sphere S^2 consists of the north pole N.

More generally, the cut locus of a point $p \in S^n$ is the singleton $\{-p\}$





The cut locus of the equatorial S^1 in the sphere S^2 is the set $\{N, S\}$, which is S^0.

More generally, the cut locus of $S^k \subset S^n$ is S^{n-k-1}

The cut locus of $\mathbb{RP}^k \subset \mathbb{RP}^n$ is $\mathbb{RP}^{n-k-1}$, and the cut locus of $\mathbb{CP}^k \subset \mathbb{CP}^n$ is $\mathbb{CP}^{n-k-1}$.

The cut locus of $O(n, \mathbb{R})$ in $M(n, \mathbb{R})$ is $M(n, \mathbb{R}) \setminus \mathrm{GL}(n, \mathbb{R})$, the set of singular matrices (Basu-Prasad [2023][†]). This gives a geometric interpretation of the Gram-Schmidt process.

[†] Somnath Basu and Sachchidanand Prasad. *A connection between cut locus, Thom space and Morse-Bott functions.* Algebr. Geom. Topol. (2023).

Cut Time and Injectivity Radius

Given a unit normal vector $\mathbf{n} \in TN^{\perp_{\mathfrak{g}}}$, the cut time is defined as

$$\rho_{\mathfrak{g}}(N, \mathbf{n}) := \sup\{t \mid \operatorname{dist}_{\mathfrak{g}}(N, \gamma^{\mathfrak{g}}_{\mathbf{n}}(t)) = t\},$$

where $\gamma^{\mathfrak{g}}_{\mathbf{n}}$ is the unique geodesic w.r.t. $\mathfrak{g}$, with initial velocity $\mathbf{n}$.

Given $N \subset M$, the injectivity radius of N is defined as

$$\operatorname{Inj}(N, \mathfrak{g}) := \operatorname{dist}_{\mathfrak{g}}(N, \operatorname{Cut}(N, \mathfrak{g})).$$

The injectivity radius of $(M, \mathfrak{g})$ is defined as

$$\operatorname{Inj}(\mathfrak{g}) := \inf_{p \in M} \operatorname{Inj}(p, \mathfrak{g}).$$

We have the following result, originally proved by Ehrlich, and later simplified by Sakai.

Theorem : (Ehrlich [1974][†], Sakai [1983][‡])

Suppose $\mathfrak{g}_i$ is a sequence of Riemannian metrics on a closed manifold M, converging to the metric $\mathfrak{g}_0$ in the Whitney C^2 topology. Then, $\lim_i \operatorname{Inj}(\mathfrak{g}_i) = \operatorname{Inj}(\mathfrak{g}_0)$. In other words, $\mathfrak{g} \mapsto \operatorname{Inj}(\mathfrak{g})$ is continuous.

Remark

Whitney C^k topology is given by the uniformly on compact convergence of the kth jet data; since M is compact we have the strong and the weak Whitney topologies coincide, which is, moreover, metrizable.

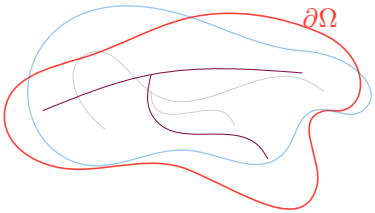
[†] Paul E. Ehrlich. *Continuity properties of the injectivity radius function.* Compositio Math. (1974).

[‡] Takashi Sakai. *On continuity of injectivity radius function.* Math. J. Okayama Univ. (1983).

Stability of the Cut Locus

The stability of the cut locus is well-studied in the literature from different perspectives.

- **Buchner (1977)[†]:** $\mathrm{Cut}(p, \mathfrak{g})$ is stable from the point of view of singularity theory for generic $\mathfrak{g}$, on a manifold of dimension ≤ 6.



Cut locus of $\partial\Omega$ inside Ω

- **Ardoy-Guijarro (2011)[‡]:** For an open set $\Omega \subset M$ with boundary $\partial\Omega$, $\mathrm{Sep}(\partial\Omega, \mathfrak{g}) \cap \Omega$ is Hausdorff stable when the boundary $\partial\Omega$ is perturbed by C^2-small diffeomorphisms.
- **Albano (2016)[§]:** $\mathrm{Cut}(\partial\Omega, \mathfrak{g})$ is Hausdorff stable, when both the boundary $\partial\Omega$ of a bounded domain $\Omega \subset \mathbb{R}^n$, and the metric are perturbed in a C^2-small sense.

- **Eltzner-García-Huckemann-Tuschmann (2021)[⁕]:** Cut locus is topologically stable from the point of view of metric measure theory.

In recent work, we have adapted the approach of Albano [2016][§].

[†] Michael A. Buchner. *Stability of the cut locus in dimensions less than or equal to* 6. `Invent. Math.` (1977).

[‡] Pablo Angulo Ardoy and Luis Guijarro. *Cut and singular loci up to codimension* 3. `Ann. Inst. Fourier (Grenoble).` (2011).

[§] Paolo Albano. *On the stability of the cut locus.* `Nonlinear Anal.` (2016).

[⁕] Benjamin Eltzner, Fernando Galaz-García, Stephan Huckemann, and Wilderich Tuschmann. *Stability of the cut locus and a central limit theorem for Frechet means of Riemannian manifolds.* `Proc. Amer. Math. Soc.` (2021).

Cut Locus as a Map

Given closed sets $A, B \subset M$ of a metric space (M, d), the Hausdorff distance between them is

$$d_H(A, B) := \max\left\{\sup_{a\in A} \operatorname{dist}(a, B), \quad \sup_{b\in B} \operatorname{dist}(A, b)\right\}.$$

Let M be a compact manifold without boundary, and $\mathfrak{h}$ be an auxiliary metric on M.

- Let $\mathcal{G} = \mathcal{G}(M)$ be the collection of C^2 Riemannian metrics on M, equipped with the Whitney C^2-topology.
- Let $\mathcal{K} = \mathcal{K}(M) := \{K \subset M \mid K \text{ is compact}\}$, equipped with the topology induced by the Hausdorff metric d_H associated with $\mathfrak{h}$. The topology is independent of the choice of $\mathfrak{h}$.
- Given a manifold N, let $\operatorname{Emb}(N, M)$ be the space of smooth embeddings $N \hookrightarrow M$, equipped with the Whitney C^2-topology.

As the cut locus is closed, we have the maps

$$\begin{aligned} \operatorname{Cut} : M \times \mathcal{G} &\to \mathcal{K} \\ (p, \mathfrak{g}) &\mapsto \operatorname{Cut}(p, \mathfrak{g}), \end{aligned} \qquad \begin{aligned} \operatorname{Cut} : \operatorname{Emb}(N, M) \times \mathcal{G} &\to \mathcal{K} \\ (f, \mathfrak{g}) &\mapsto \operatorname{Cut}(f(N), \mathfrak{g}), \end{aligned}$$

We also have the map

$$\begin{aligned} \operatorname{Inj} : \operatorname{Emb}(N, M) \times \mathcal{G} &\to (0, \infty) \\ (f, \mathfrak{g}) &\mapsto \operatorname{Inj}(f(N), \mathfrak{g}). \end{aligned}$$

Stability of Cut Locus and Injectivity Radius

Theorem : (_, Itoh, Prasad [2026][†])

The cut locus map $\mathrm{Cut} : M \times \mathcal{G} \to \mathcal{K}$ is continuous.

Recently, we have generalized the result of Sakai [1983].

Theorem : (_ [2026][‡])

Given closed manifolds N, M, the injectivity radius map

$$\begin{aligned} \mathrm{Inj} : \mathrm{Emb}(N, M) \times \mathcal{G} &\to (0, \infty) \\ (f, \mathfrak{g}) &\mapsto \mathrm{Inj}(f(N), \mathfrak{g}). \end{aligned}$$

is continuous.

As a consequence, we get the following.

Theorem : (_ [2026][‡])

Given a closed manifold N, the cut locus map $\mathrm{Cut} : \mathrm{Emb}(N, M) \times \mathcal{G} \to \mathcal{K}$ is continuous.

As the spaces involved are metrizable, we only need to verify sequential continuity.

[†] **Aritra Bhowmick**, Jin-ichi Itoh and Sachchidanand Prasad. *Hausdorff stability of the cut locus under C^2-perturbations of the metric.* J. Math. Anal. Appl. (2026).

[‡] **Aritra Bhowmick**. *Stability of the Injectivity Radius and the Cut Locus of Submanifolds under Perturbations.* Comm. Anal. Geom. (to appear) (2026).

Proof Sketch of Stability: Ingredients

We have the characterization of convergence in the Hausdorff metric.

Lemma

Suppose M is compact, $X_i, X \subset M$ are closed (and hence compact) sets. Then, $\lim_i d_H(X_i, X) = 0$ if and only if the following holds:

- for any $x \in X$, there exists $x_i \in X_i$ with $\lim_i x_i = x$, and
- for any convergent sequence $x_{i_j} \in X_{i_j}$, we have $\lim_j x_{i_j} \in X$.

We also have a useful characterization of a separating point. Given $\Omega \subset M$ open, consider the cut locus (resp. separating set) of the boundary $\partial\Omega$ inside Ω:

$$\mathrm{Cut}^{\mathrm{in}}(\Omega, \mathfrak{g}) := \mathrm{Cut}(\partial\Omega, \mathfrak{g}) \cap \Omega, \qquad \mathrm{Sep}^{\mathrm{in}}(\Omega, \mathfrak{g}) := \mathrm{Sep}(\partial\Omega, \mathfrak{g}) \cap \Omega.$$

The distance function $u := \mathrm{dist}_{\mathfrak{g}}(\partial\Omega, _)$ is *not* C^1 on points of $\mathrm{Sep}^{\mathrm{in}}(\Omega, \mathfrak{g})$, and the C^1 singular support of u inside Ω is precisely $\mathrm{Cut}^{\mathrm{in}}(\Omega, \mathfrak{g})$.

Lemma

$x \in \mathrm{Sep}^{\mathrm{in}}(\Omega, \mathfrak{g})$ if and only if there exists a C^1 function φ near x such that $u - \varphi$ attains a (strict) local maximum at x and $\|d_x\varphi\|_{\mathfrak{g}} < 1$.

The characterization is a consequence of the fact that u is the unique (positive) viscosity solution of the eikonal equation $\|\nabla f\|_{\mathfrak{g}} = 1$ with $f|_{\partial\Omega} = 0$.

Proof Sketch of Stability

Lemma

Suppose $\mathfrak{g}_i \to \mathfrak{g}_0$ is a C^2-convergent sequence of metrics. Given $x \in \mathrm{Cut}(N, \mathfrak{g}_0)$, there exists a sequence $x_i \in \mathrm{Cut}(N, \mathfrak{g}_i)$ such that $\lim_i x_i = x$.

Proof : Since $\mathrm{Cut}(N, \mathfrak{g}_0) = \overline{\mathrm{Sep}(N, \mathfrak{g}_0)}$, assume $x \in \mathrm{Sep}(N, \mathfrak{g}_0)$. As $\lim_i \mathrm{Inj}(N, \mathfrak{g}_i) = \mathrm{Inj}(N, \mathfrak{g}_0) > 0$, we have $\delta > 0$ such that $\delta < \mathrm{Inj}(N, \mathfrak{g}_i)$. Fix the sets and the distance functions:

$$\Omega_i := \left\{ x \in M \;\middle|\; \mathrm{dist}_{\mathfrak{g}_i}(N, x) > \delta \right\}, \qquad u_i := \mathrm{dist}_{\mathfrak{g}_i}(\partial\Omega_i, _) = \mathrm{dist}_{\mathfrak{g}_i}(N, _) - \delta.$$

As $\mathfrak{g}_i \overset{C^2}{\to} \mathfrak{g}_0$, we have $u_i \to u_0$ uniformly on compacts. As $x \in \mathrm{Sep}(N, \mathfrak{g}_0) = \mathrm{Sep}^{\mathrm{in}}(N, \mathfrak{g}_0)$, we have a C^1 function φ near x such that $u - \varphi$ attains a strict local maxima at x, and

$$\left\| d_x \varphi \right\|_{\mathfrak{g}_0} < 1.$$

As $u_i \to u_0$ uniformly, we have a sequence x_i with $\lim_i x_i = x$, such that $u_i - \varphi$ attains a local maxima at x_i. As u_i is a viscosity (sub)solution, we have $\left\| d_{x_i} \varphi \right\|_{\mathfrak{g}_i} \leq 1$. Now,

$$\lim_i \left\| d_{x_i} \varphi \right\|_{\mathfrak{g}_i} = \left\| d_x \varphi \right\|_{\mathfrak{g}_0} < 1.$$

Hence, $\left\| d_{x_i} \varphi \right\|_{\mathfrak{g}_i} < 1$ for $i \gg 1$. But then $x_i \in \mathrm{Sep}^{\mathrm{in}}(\Omega_i, \mathfrak{g}_i) = \mathrm{Sep}(N, \mathfrak{g}_i)$, concluding the proof. □

Proof Sketch of Stability (cont.)

Lemma

Suppose $\mathfrak{g}_i \to \mathfrak{g}_0$ is a C^2-convergent sequence of metrics. Given $x_{i_j} \in \mathrm{Cut}\left(N, \mathfrak{g}_{i_j}\right)$ with $x = \lim_j x_{i_j}$, we have $x \in \mathrm{Cut}(N, \mathfrak{g}_0)$.

Proof of the Continuity : The proof of the continuity of the cut locus map now follows from the previous two lemmas, and the characterization of the convergence in the Hausdorff metric. □

Corollary

Suppose $\mathfrak{g}_i \to \mathfrak{g}_0$ is a C^2-convergent sequence of metrics, and $f_i \to f_0$ is a C^2-convergent sequence of embeddings $N \hookrightarrow M$; set $N_i = f_i(N)$. Let $\mathbf{n}_i \in TN_i^{\perp_{\mathfrak{g}_i}}$ be unit normal vectors converging to some $\mathbf{n}_0$, necessarily in $TN_0^{\perp_{\mathfrak{g}_0}}$ with $\|\mathbf{n}_0\|_{\mathfrak{g}_0} = 1$. Then, $\lim_i \rho_{\mathfrak{g}_i}(N_i, \mathbf{n}_i) = \rho_{\mathfrak{g}_0}(N_0, \mathbf{n}_0)$.

The above strengthens the classical result about the continuity of the cut time map of a fixed submanifold, with respect to a fixed metric.

Focal Points

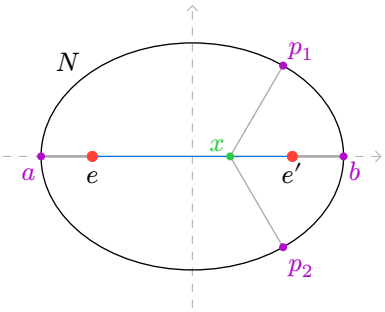
A cut point may not always be a separating point.

Let N be an ellipse in $\mathbb{R}^2$, with focal points e, e'.

$$\mathrm{Sep}(N) = \textit{Open} \text{ interval between } e \text{ and } e'.$$

For some x on the open interval, we have $[p_1, x]$ and $[p_2, x]$ as N-segments. For e (resp. e'), we have *unique* N-segment $[a, e]$ (resp. $[b, e']$), any extension of the of which fails to be distance minimizing from N. Thus,

$$\mathrm{Cut}(N) = \textit{Closed} \text{ interval joining } e \text{ and } e'.$$



Definition

Given $N \hookrightarrow M$, a point $q \in M$ is called a focal point of N if it is a *critical value* of the normal exponential map $\exp^{\perp} : TN^{\perp} \to M$

Theorem : (_, Prasad [2024][†])

For any $q \in \mathrm{Cut}(N)$, we have a *non-exclusive* dichotomy:

$q \in \mathrm{Sep}(N)$, or, q is the *first* focal point of N along some N-segment.

[†] **Aritra Bhowmick** and Sachchidanand Prasad. *On the cut locus of submanifolds of a Finsler manifold.* J. Geom. Anal. (2024).

Intersection of Cut and Focal Loci

In 1959, Rauch made the following conjecture.

Conjecture : (Rauch [1959][†])

For any point p in a compact, simply connected, Riemannian manifold $(M, \mathfrak{g})$, the cut locus $\mathrm{Cut}(p, \mathfrak{g})$ must have a conjugate point of p along some distance-minimizing geodesic from p.

The conjecture is true for S^2, and for symmetric spaces.

Theorem : (Weinstein [1968][‡])

If M is a compact manifold without boundary, not homeomorphic to S^2, then there exists a metric $\mathfrak{g}$ on M and a point $p \in M$, such that the cut locus $\mathrm{Cut}(p, \mathfrak{g})$ has no conjugate point of p along any distance-minimizing geodesic from p.

We have generalized this to the case of submanifolds.

Theorem : (_, Prasad, Schick [§])

Suppose M is a compact, connected, manifold without boundary, with $\dim M \geq 3$. Let $N \subset M$ be a closed, embedded submanifold, each component being without boundary, and having codimension ≥ 1. Then, there exists a metric $\mathfrak{g}$ on M such that the cut locus $\mathrm{Cut}(N, \mathfrak{g})$ contains no focal point of N along an N-segment.

[†] Harry Ernest Rauch. *Geodesics and curvature in differential geometry in the large.* (1959).

[‡] Alan D. Weinstein. *The cut locus and conjugate locus of a riemannian manifold.* `Ann. of Math.` (1968).

[§] **Aritra Bhowmick**, Sachchidanand Prasad and Thomas Schick. *Intersection of Cut and Focal Loci of a Submanifold.* (in preparation).

An Estimate on the First Focal Time

Principal curvature : generalization of the curvature of a planar curve.

Sectional curvature : generalization of the Gaussian curvature of a surface.

Theorem : (Warner [1966][†])

Let $H \subset M$ be a codimension 1 submanifold in a compact Riemannian manifold $(M, \mathfrak{g})$, and $\mathbf{n}$ be a unit vector field normal to H at every point. Suppose $\Delta_H, \kappa_M > 0$ are real numbers such that

- the principal curvatures of H along $\mathbf{n}$ are $> -\Delta_H$, and
- all the sectional curvatures of M are $< \kappa_M$.

Set

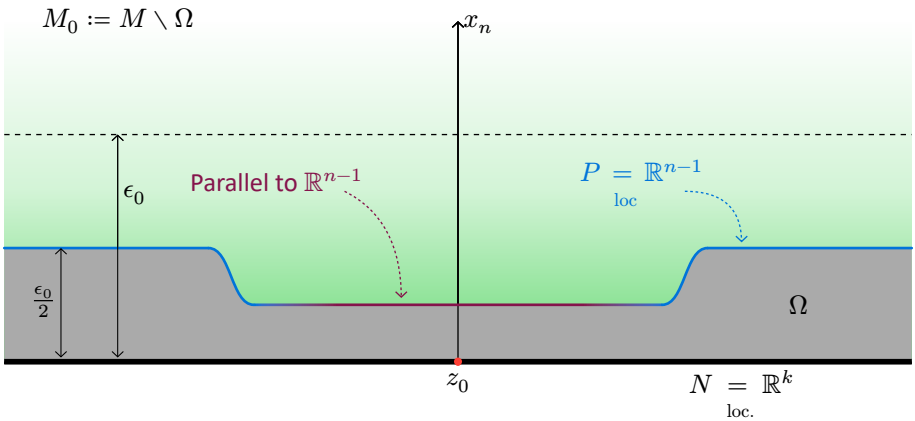
$$\varepsilon_H := \frac{1}{\sqrt{\kappa_M}} \arctan\left(\frac{\sqrt{\kappa_M}}{\Delta_H}\right).$$

Then, for any unit speed geodesic $\gamma : [0, \infty) \to M$ in the direction of $\mathbf{n}$, there is no focal point of H along γ in the interval $[0, \varepsilon_H]$.

[†]Frank Warner. *Extensions of the Rauch comparison theorem to submanifolds.* Trans. Amer. Math. Soc. (1966).

Step 1: Choosing an Initial Metric and a Tubular Neighborhood

For $z_0 \in N$, get a metric which is Euclidean near z_0 and N is isometric to $\mathbb{R}^k \times \{0\} \subset \mathbb{R}^n$. Choose an ϵ_0-tubular neighborhood of N, which is $\mathbb{R}^k \times \mathbb{D}^{n-k}_{\epsilon_0}$ near z_0. Get a smaller tubular neighborhood Ω, whose boundary P is flat and parallel to $\mathbb{R}^{n-1} \times \{0\}$ near z_0.



$M_0 := M \setminus \Omega$ is a compact, connected, codimension 0 submanifold, with boundary P

Step 2: Constructing a Dense Curve with Bounded Curvature

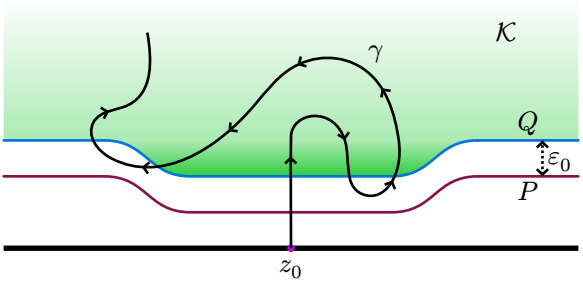
Theorem : (_, Prasad, Schick)

There exists $\Delta_{\text{curve}} > 0$, such that we can get a regular curve in $M_0 \setminus P$, with geodesic curvature strictly bounded by Δ_{curve}, which is arbitrarily dense in a given compact set $\mathcal{K} \subset M_0 \setminus P$. The curve can be assumed to be embedded if $\dim M_0 \geq 3$.

Fix $\Delta_P, \kappa_M > 0$, such that principal curvatures of P are $> -\Delta_P$ along the outward normal directions, and sectional curvatures of M are $< \kappa_M$. Set,

$$\Delta_{\text{final}} := \max\left\{\sqrt{\frac{32}{27}}\Delta_{\text{curve}},\quad \Delta_P\right\}, \qquad \varepsilon_0 := \frac{1}{\sqrt{\kappa_M}}\arctan\left(\frac{\sqrt{\kappa_M}}{\Delta_{\text{final}}}\right).$$

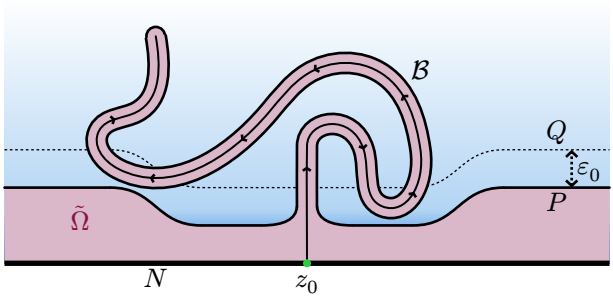
Get ε_0-neighborhood of P, and set $\mathcal{K}$ as the region away from N, with boundary Q.



Get embedded regular curve γ, $\frac{\varepsilon_0}{2}$-dense in $\mathcal{K}$, with strict geodesic curvature bound Δ_{curve}.

Step 3: Constructing a New Tubular Neighborhood

Next, we get a tube around the curve γ, and join the boundary smoothly to P.



The new region $\tilde{\Omega}$ is a tubular neighborhood of N, with boundary $\mathcal{B}$. We estimate the principal curvatures of $\mathcal{B}$.

- By an argument of Weinstein, around γ, the principal curvatures of $\mathcal{B}$ along the outward normal directions are $> -\sqrt{\frac{32}{27}}\Delta_{\text{curve}} \geq -\Delta_{\text{final}}$.
- On P, the bound is $> -\Delta_P \geq -\Delta_{\text{final}}$.
- The funnel-shaped join is in a Euclidean region, where we can perform explicit computations.

By Warner's theorem: there are no focal points of $\mathcal{B}$ in the outward normal direction for time $[0, \varepsilon_0)$.

Hence, by the $\frac{\varepsilon_0}{2}$-denseness, a cut point of $\mathcal{B}$ on the outside $M_0 \setminus \tilde{\Omega}$ is *not* a focal point of $\mathcal{B}$ along any $\mathcal{B}$-segment.

Step 4: Changing the Metric inside the Tubular Neighborhood

Theorem : (_, Prasad, Schick)

There exists a new metric $\mathfrak{g}$ on M such that the following holds.

- $\mathfrak{g}$ is unchanged on the complement of the tubular neighborhood $\tilde{\Omega}$, near its boundary $\mathcal{B}$, and near the submanifold N.
- $\tilde{\Omega}$ is a *unit normal neighborhood* of N with respect to $\mathfrak{g}$.

Proof of Main Theorem : With the new metric $\mathfrak{g}$,

- every $\mathcal{B}$-segment $\gamma : [0, \epsilon] \to M_0 \setminus \tilde{\Omega}$ extends back to an N-segment $\tilde{\gamma} : [-1, \epsilon] \to M$, and
- every N-segment $\eta : [0, 1] \to M$ extends to an $\mathcal{B}$-segment $\tilde{\eta} : [0, 1 + \tau] \to M$ for some $\tau > 0$.

Consequently,

- cut points of N are precisely the cut points of $\mathcal{B}$ in $M_0 \setminus \tilde{\Omega}$, and
- focal points of N and focal points of $\mathcal{B}$ in $M_0 \setminus \tilde{\Omega}$ match, up to a time-shift of 1.

Hence, with respect to $\mathfrak{g}$, cut locus of N is free of focal points of N along N-segments. □

Some Consequences

Theorem : (_, Prasad, Schick)

Let $N_1, N_2 \subset M$ be compact, connected, disjointly embedded, submanifolds, without boundary. Then, the following are equivalent.

- There is a metric $\mathfrak{g}$ on M such that $N_2 = \mathrm{Cut}(N_1, \mathfrak{g})$.
- M is diffeomorphic to the union of two disc bundles over N_1 and N_2, attached along the boundary by a diffeomorphism.
- There exists a Morse-Bott function with exactly two critical submanifolds N_1 and N_2.
- There exists a function $f : M \to \mathbb{R}$ with critical set $N_1 \sqcup N_2$.

Corollary : (B.-Prasad-Schick, Basu-Prasad [2026][†], Lerario-Meroni-Zuddas [2024][‡])

Given a closed manifold M, if there exists a function $f : M \to \mathbb{R}$ with exactly two critical values a, b, and if $S^p = f^{-1}(a), S^q = f^{-1}(b)$, with $p \neq q$, then M is homeomorphic to S^{p+q+1}.

Corollary : (_ [2026][§])

Let $(M, \mathfrak{h})$ be a closed Riemannian manifold. Suppose $N \subset M$ is a closed submanifold such that $\mathrm{Cut}(N, \mathfrak{h})$ is free of focal points of N along N-segments. Then, there exists a metric $\mathfrak{g}$ on M such that $\mathrm{Cut}(N, \mathfrak{g})$ is a finite subcomplex of M (and still free of focal points).

[‡] Somnath Basu and Sachchidanand Prasad. *On manifolds homeomorphic to spheres* (preprint) (2026).

[†] Antonio Lerario, Chiara Meroni, and Daniele Zuddas. *On smooth functions with two critical values.* Rev. Mat. Complut. (2024).

[§] **Aritra Bhowmick**. *Stability of the Injectivity Radius and the Cut Locus of Submanifolds under Perturbations.* Comm. Anal. Geom. (to appear) (2026).

Ongoing Work

Finsler geometry is a generalization of Riemannian geometry, where one can only measure angles.

Lorentzian geometry comes up in the study of space-time.

Both of them can be generalized under the framework of wind Finsler geometry.

Modeling Wildfire: Forrest wildfires are modeled via methods in Riemannian geometry; fire is concentrated at *focal points* of the front of the fire.

Recent works of M. A. Javaloyes and his collaborators have shown the potential of modeling wildfire using wind Finsler geometry, where one can take into account both wind direction and slope of the ground.

Our immediate goals are the following.

- Can we extend the joint work with S. Prasad[†] to the setup of wind Finsler geometry?
- Can the stability of the cut locus be proved in the context of Lorentzian geometry?
- Under the assumption that the cut locus is a finite subcomplex, can we bring some algebraic topology machinery into play?

[†] **Aritra Bhowmick** and Sachchidanand Prasad. *On the cut locus of submanifolds of a Finsler manifold.* J. Geom. Anal. (2024).

Thank you!